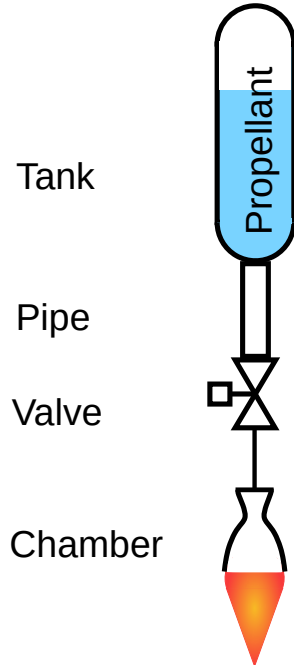


State Space Model

Monopropellant Pressure Fed



ODEs

$$\dot{p}_p = \frac{a^2}{V} (\dot{m}_p - \dot{m}_v)$$

$$\ddot{m}_p = \frac{A}{l} \left(p_{tank} - p_p - \frac{\zeta}{\rho} \dot{m}_p |\dot{m}_p| \right)$$

$$\dot{k}_v = \frac{k_{v,set} - k_v}{T}$$

$$\dot{p}_c = \frac{RT}{V} \left(\dot{m}_v - \frac{p_c A_{th}}{c^*} \right)$$

Algebraic Relations

$$\zeta = \lambda \frac{l}{d} \frac{1}{2A^2}$$

$$\frac{1}{\sqrt{\lambda}} = -2 \lg \left(\frac{2.51}{Re \sqrt{\lambda}} \right)$$

$$\dot{m}_{valve} = \sqrt{\frac{|\Delta p|}{3600^2 \cdot 1e5}} \rho k_v \text{sign}(\Delta p)$$

$$\Delta p = p_p - p_c$$

$$F = \dot{m}_c c_e = \frac{p_c A_{th}}{c^*} \cdot c_e$$